

Stage 1: Supervised Fine-Tuning (SFT)

Thinking Traces

- Crystallographic Symmetry
- Local Coordination Environments
- Predicted Functional Properties

Progressive Reasoning:
Abstract Symmetries -> Local environments -> Concrete Properties

Stage 2: Reinforcement Learning (RL)

Multi-Object Dense Reward

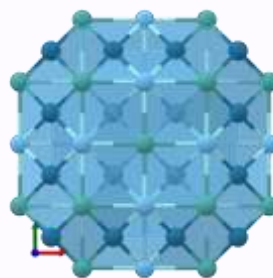
- Physical Validity
- Chemical Validity
- Thermodynamic Stability

Task-Specific Rewards

- Discrete: space group match
- Continuous: Elasticity, Thermal Expansion, ...

Validity and Stability Optimization

Below is a description of a bulk material. The **chemical formula** is TiMoRe2. The energy above the convex hull is 0.0. The formation energy per atom is -0.3529. The band gap is 0.0. The **spacegroup** number is 225.



Conditioning on Properties

Below is a description of a bulk material. The **chemical formula** is TiMoB2. The **bulk_modulus** is in [150, 300] GPa. The **shear_modulus** is greater than 200 GPa.

